

WELL COMPLETION REPORT

Synthetic Field Cluster — SYN-Assam-Duliajan (Upper-Assam-type setting)

1. WELL IDENTIFICATION

Well ID	W004
Well Name	SYN-Assam-Duliajan-004
Field / Cluster	SYN-Assam-Duliajan
Area	Synthetic Assam Field Cluster (Duliajan-type operating area)
Latitude	27.412500 N
Longitude	95.349472 E
Well Status	Drilled
Well Role	Historical (offset reference well)
Spud Date	28-Oct-2008
Drilling Completed	08-Nov-2008
Report / Completion Date	12-Nov-2008
Total Depth (Driller)	3,590 m
Primary Formation	Sandstone
Formations Penetrated	Claystone / Shale / Sandstone / Limestone
Rig	SYN-RIG-E1400-04
Drilling Days	12 days
Recorded Events	2
Recorded NPT	14.2 hours

2. MACHINE-READABLE HEADER BLOCK

Document Type: Well Completion Report
Well ID: W004
Well Name: SYN-Assam-Duliajan-004
Field: SYN-Assam-Duliajan
Latitude: 27.412500
Longitude: 95.349472
Total Depth: 3,590 m
Primary Formation: Sandstone
Well Status: Drilled
Spud Date: 2008-10-28
Event Count: 2
Total NPT: 14.2 hours

PROTOTYPE TRAINING RECORD. Synthetic demonstration document prepared for the eRTMAC-NWIS prototype (SIH PS 26121). The well, coordinates, formations, events and measurements described here are generated for software demonstration and are not an operational record of any company. Location names are used for interface readability only.

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3. WELL SUMMARY

3.1 Objective

Well W004 (SYN-Assam-Duliajan-004) was drilled to drill a development location targeting the primary sandstone objective and obtain a complete formation and event record for offset reference. The well was drilled vertically from surface to a driller's total depth of 3,590 m within the SYN-Assam-Duliajan synthetic cluster.

3.2 Drilling Interval

Surface to 3,590 m, completed in 12 rotary drilling days between 28-Oct-2008 and 08-Nov-2008. Average overall penetration rate for the well was 12.5 m/hr on a 24-hour basis.

3.3 Formations Encountered

The well penetrated 4 formation units: Claystone (0–822 m), Shale (822–2,009 m), Sandstone (2,009–3,180 m), Limestone (3,180–3,590 m). The primary objective formation was Sandstone.

3.4 Major Operational Events

A total of 2 reportable events were recorded, comprising 2 × Mud Loss. Aggregate non-productive time was 14.2 hours. Event detail is given in Section 6.

3.5 Casing Operations

3 casing strings were run: 13-3/8" at 820 m, 9-5/8" at 2,010 m, 7" at 3,590 m. All strings were cemented as detailed in Section 7.

3.6 Completion Status

Well status at report date: **Drilled**. The well was drilled and logged to total depth and suspended pending completion decisions.

4. OFFSET-WELL CONTEXT

The following synthetic wells of the same cluster lie within short offset distance and share a comparable formation sequence. They are the primary comparison set for offset-well analysis.

Well ID	Well Name	Distance (km)	TD (m)	Primary Formation	Events
W001	SYN-Assam-Duliajan-001	4.92	3,470	Sandstone	4
W009	SYN-Assam-Duliajan-009	5.01	3,180	Sandstone	6
W007	SYN-Assam-Duliajan-007	5.79	3,400	Sandstone	3
W002	SYN-Assam-Duliajan-002	6.54	3,450	Sandstone	7
W006	SYN-Assam-Duliajan-006	8.40	3,960	Sandstone	0
W003	SYN-Assam-Duliajan-003	9.47	3,850	Sandstone	8

5. FORMATION / DEPTH RECORD

Formation tops and bases as recorded from cuttings and log correlation. Depths are driller's depths below rotary table.

Unit	Formation	Top Depth (m)	Bottom Depth (m)	Thickness (m)	Remarks
1	Claystone	0	822	822	Routine drilling, no reportable event
2	Shale	822	2,009	1,187	Routine drilling, no reportable event
3	Sandstone	2,009	3,180	1,171	Mud Loss recorded (1 event)
4	Limestone	3,180	3,590	410	Mud Loss recorded (1 event)

Formation: Claystone Top: 0 m Bottom: 822 m
Formation: Shale Top: 822 m Bottom: 2009 m
Formation: Sandstone Top: 2009 m Bottom: 3180 m
Formation: Limestone Top: 3180 m Bottom: 3590 m

6. DRILLING PARAMETERS BY FORMATION

Representative values summarised from the depth-indexed parameter record for this well. Mud weight is reported in ppg, rate of penetration in m/hr, rotary speed in rpm and weight on bit in klbf.

Formation	Interval (m)	Mud Wt (ppg)	MW range	ROP (m/hr)	RPM	WOB (klbf)
Claystone	0–822	9.59	9.0–10.1	31.7	140	8.3
Shale	822–2,009	11.11	10.3–12.0	16.2	124	17.9
Sandstone	2,009–3,180	12.60	11.7–13.4	15.6	112	25.7
Limestone	3,180–3,590	14.02	13.7–14.3	7.4	90	35.6

6.1 Mud Programme Summary

Water-based inhibited mud was used throughout. Mud weight was raised progressively with depth from approximately 9.6 ppg in the upper section to 14.0 ppg at total depth. Mud weight was trimmed to the minimum safe value across the recorded loss intervals and lost-circulation material was maintained in the active system.

6.2 Bit / Hydraulics Note

Rotary speed was held in the 120 rpm range on average with weight on bit averaging 20.3 klbf. Penetration rate declined with depth from 31.7 m/hr in the upper hole to 7.4 m/hr at total depth, consistent with the area trend.

7. DRILLING EVENTS RECORD

All reportable events for this well, in depth order. Each event is recorded against the formation interval containing the event depth.

Event ID	Depth (m)	Formation	Event Type	Severity	Duration (hr)
E0004	2,865	Sandstone	Mud Loss	Medium	6.8
E0034	3,237	Limestone	Mud Loss	Medium	7.4

Well ID: W004	Event Type: Mud Loss	Depth: 2865 m	Formation: Sandstone	Severity: Medium	Duration: 6.8 hours
Well ID: W004	Event Type: Mud Loss	Depth: 3237 m	Formation: Limestone	Severity: Medium	Duration: 7.4 hours

7.1 EVENT DETAIL

EVENT E0004 — MUD LOSS AT 2,865 m (SANDSTONE)
Well ID: W004 Event ID: E0004 Event Type: Mud Loss Depth: 2865 m Formation: Sandstone Severity: Medium Duration: 6.8 hours Operational Impact: Partial non-productive time; drilling suspended while the well was treated and conditioned. Description. Partial loss of drilling fluid recorded at 2865 m in the Sandstone interval; returns reduced and LCM pumped to regain circulation. Parameters recorded around the event depth: mud weight 12.89 ppg, ROP 15.2 m/hr, 105 rpm, WOB 27.7 klb. Response. Pumps slowed and a lost-circulation-material pill (fibrous/granular blend) spotted across the loss zone; mud weight trimmed to the minimum safe value and ECD reduced; pit levels monitored on a dedicated loss sheet. Outcome. Returns re-established and losses reduced to seepage. Drilling resumed with reduced flow rate and controlled ROP through the remainder of the interval. Historical observation. Loss-zone behaviour is depth- and formation-controlled in this area; the porous sand intervals accept fluid once ECD exceeds the fracture/pore window.
EVENT E0034 — MUD LOSS AT 3,237 m (LIMESTONE)
Well ID: W004 Event ID: E0034 Event Type: Mud Loss Depth: 3237 m Formation: Limestone Severity: Medium Duration: 7.4 hours Operational Impact: Partial non-productive time; drilling suspended while the well was treated and conditioned. Description. Loss of circulation into porous Limestone at 3237 m; drilling paused and a lost-circulation-material pill spotted. Parameters recorded around the event depth: mud weight 13.73 ppg, ROP 8.1 m/hr, 91 rpm, WOB 34.2 klb. Response. Pumps slowed and a lost-circulation-material pill (fibrous/granular blend) spotted across the loss zone; mud weight trimmed to the minimum safe value and ECD reduced; pit levels monitored on a dedicated loss sheet. Outcome. Returns re-established and losses reduced to seepage. Drilling resumed with reduced flow rate and controlled ROP through the remainder of the interval. Historical observation. Loss-zone behaviour is depth- and formation-controlled in this area; the porous sand intervals accept fluid once ECD exceeds the fracture/pore window.

8. CASING AND CEMENTING RECORD

Casing	Grade / Weight	Shoe Depth (m)	Hole Size	Cement Programme
13-3/8"	K-55, 68 lb/ft	820	17-1/2"	Lead 12.6 ppg / Tail 15.8 ppg, cement to surface
9-5/8"	N-80, 47 lb/ft	2,010	12-1/4"	Lead 12.9 ppg / Tail 15.8 ppg, TOC 700 m
7"	N-80, 29 lb/ft	3,590	8-1/2"	Tail 15.8 ppg, TOC 1,860 m

8.1 Cementing Observations

All casing strings were run and landed without restriction. Cement returns and bond logs were within acceptance criteria for every stage; no remedial cementing was required.

8.2 Pressure Testing

Each string was pressure tested after landing and formation integrity tests were carried out below the 13-3/8" and 9-5/8" shoes. Recorded results were satisfactory except where noted in Section 7.

9. FINAL SUMMARY

Final Depth	3,590 m
Primary Formation	Sandstone
Major Events	0 of 2 events rated High or Critical
Total NPT	14.2 hours
Completion Status	Drilled
Offset Risk Classification	MODERATE — isolated / low-severity events

9.1 Drilling Challenges

- **Mud Loss** at 2,865 m, 3,237 m (Sandstone, Limestone) — 14.2 hr cumulative.

9.2 Key Historical Observations

- Loss-zone behaviour is depth- and formation-controlled in this area; the porous sand intervals accept fluid once ECD exceeds the fracture/pore window.
- Formation tops in this well correlate closely with the neighbouring wells listed in Section 4, supporting depth-based offset comparison across the cluster.

9.3 Lessons Learned

- Approach the Sandstone/Limestone interval (2,865–3,237 m) with LCM pre-mixed and the minimum safe mud weight; monitor returns continuously through this window.

10. DOCUMENT CONTROL

Document Number	WCR-W004-R0
Revision	R0
Issue Date	12-Nov-2008
Prepared By	Drilling Engineering (prototype dataset generator)
Classification	Prototype Training Record — synthetic demonstration data
Source Datasets	wells.csv, well_ formations.csv, drilling_events.csv, drilling_parameters.csv

All identifiers, coordinates, depths, formations, events and measurements in this document are synthetic and were generated for the eRTMAC-NWIS prototype. This document is not an operational record of Oil India Limited or any other operator, and no confidential information is contained in it.